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Factors Influencing the Citation of a Patent in Another Patent Modelled through REMs

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Introduction

Citation Networks is a very old analysis problem in the field of Network Science. One can want to infer a multitude of trends from the data. The data at hand for this project contains information on patent citations, detailing interactions between patents in terms of citations received and given. The goal of the project then is to make inference on the relational dynamics of the interaction between the patent citations to understand how the patents are cited over time. The dataset is available at Kaggle, however for this project we use a subset of the whole data. We will therefore answer what factors influence the citation of a patent in another patent.

Description of the Relational Event Data

The relational event data that we are considering is that of patent citations. Here, each **event** represents a citation of a patent from some specific patent. Each **non-event** represents a citation that could have been, but was not (potential citations). **Endogenous-Covariates** include the out-degree of the receiver (the citer patent), the In-degree of the receiver (the citee patent) [Both Log-Transformed], Log-transformed cumulative number of citations received by the patent.. The **Exogenous-Covariates** on the other hand include the Jaccard Similarity between patents, the Cosine Similarity in the embedding space to tell how similar the content of the patents are, the time lag of citation (telling us how likely it is a patent gets cited the older it is), Log-transformed time from the last citation event, Publication year of the receiving (citer) patent. We therefore want to model

$$\log \lambda_{sr}(t) = \underbrace{x_{sr}^{(1)}(t)\beta + x_{sr}^{(2)}(t)\beta(t) + f(x_{sr}^{(3)})}_{\text{dydic covariates, linear/time-varying/non-linear}} + \underbrace{g(x^{(4)}(t))}_{\text{global covariates}} + \underbrace{x_{sr}^{(5)}\gamma}_{\text{random effects}}$$

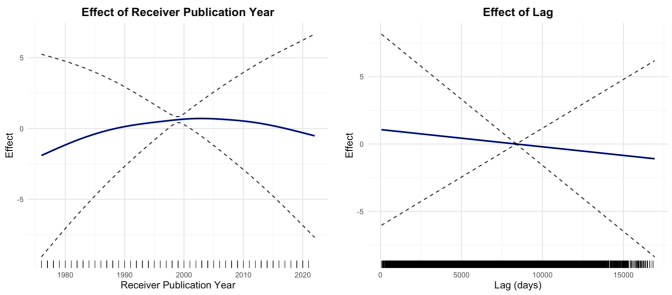
Modelling

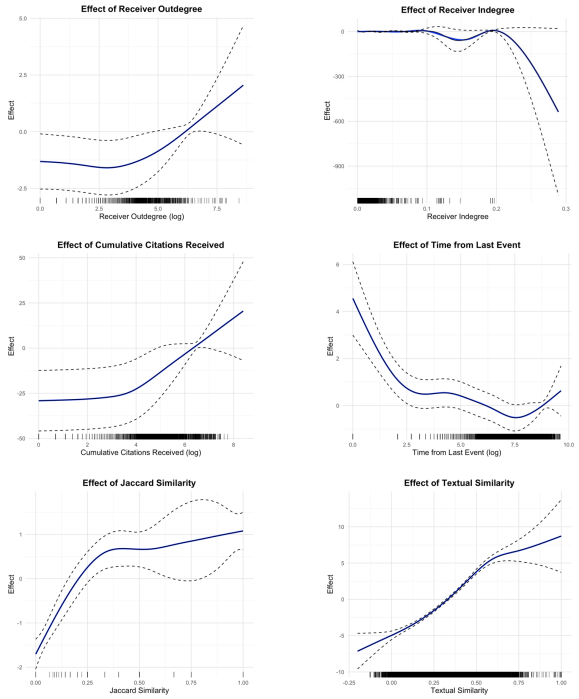
We considered two models whereby we considered all the covariates non-linearly in the **first model**. A **second model**, in which we have both the similarities measures as time-varying covariates in addition to all the non-linear covariates was also considered. We consider the covariates non-linearly as that better helps capture the interactions between them than a linear consideration would, there were no latent random effect, and global covariates considered.

Inference

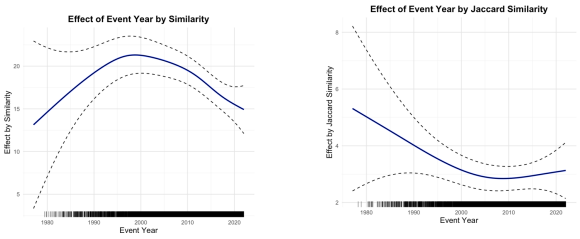
The inference procedure that we are using to compute the effects of interest as defined in the section above is conducted via the sampled partial likelihood. **Non-events** are generated by pairing patents that did not cite each other. The **Risk Set** contains all possible pairs of patents at risk (can occur) of citation at each time point 't'. The models are then compared using the AIC, and the model with less score is chosen.

Model 1 - Results





Model 2 - Results



Results and Conclusion

We have plotted the results of the effects for both the models. *Please note that the plots for the other covariates in model 2 is the same as model 1*. From looking at the summary of both the models, we can clearly say that the Effect of Lag is not significant as it has a very high p-value of 0.77. From the first plot in the first model (relevant for both), we can say that the Receiver Publication Year has had a non-linear impact on the citation, with peak around the year 2000. This suggests that patents published around this time had a higher propensity to be cited compared to other periods. From the effect of Lag, we can say that as the time lag increases, the likelihood of citation slightly decreases. From the degree of the patents, we infer that patents with higher outdegree are more likely to be cited, patents with higher indegree are less likely to be cited. We also see that patents with more citations are more likely to receive more citations. From time since last event, we can see that that recent events decrease the likelihood of additional citations. From the Jaccard Similarity measure, we can see that patents with higher similarity in technology classes are more likely to be cited, reflecting a tendency to cite similar work. Similarly, from textual similarity, we can infer that patents with higher textual similarity are more likely to be cited, indicating a preference for citing textually similar patent. In Model 2, we can see that the Jaccard Similarity over time indicates that the impact of technological class similarity on citations has diminished over the years, and for textual similarities, we see a non-linear pattern which shows that around the year 2000, more textually similar citations were more likely during this period. Finally, to compare the model, using AIC, we saw that **model 1 was better** than model 2, as it had a lower AIC score by 0.1. We can also see that both the similarity measures have the highest impact on the citation of a patent in addition to cumulative citations received.